	Raffles Institution **Physics Department** **Revision Package 3**	**Currents**	**15**

Important note: ****denotes questions that may require knowledge from later chapters in order to solve them.***

I. MCQs (ACJC – MI)

1. [CJC/H2/Prelims 2018/P1/19]

The electrical potential difference between two points in a wire carrying a current is

A the ratio of the power supplied to the current between the points.

B the force required to move a unit positive charge between the points.

C the ratio of the energy dissipated to the current between the points.

D the product of the square of the current and the resistance between two points.

2. [CJC/H2/Prelims 2018/P1/20]

An electrical source with internal resistance r is used to operate a lamp of resistance R.

What fraction of the total power is delivered to the lamp?

A $\frac{R+r}{R}$ **B** $\frac{R-r}{R}$ **C** $\frac{R}{R+r}$ **D** $\frac{R}{r}$

3. [CJC/H2/Prelims 2018/P1/21]

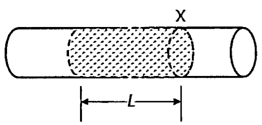
A high potential is applied between the electrodes of a hydrogen discharge tube so that the gas is ionised. Electrons then move towards the positive electrode and protons towards the negative electrode. In each second, 7×10^{18} electrons and 2×10^{18} protons pass a cross section of the tube.

The current flowing in the discharge tube is

A 0.32 A **B** 0.80 A **C** 1.12 A **D** 1.44 A

4. [EJC/H2/Prelims 2018/P1/19]

A copper wire has a number density of 8.5×10^{28} conduction electrons per cubic metre, and an cross-sectional area of 3.2×10^{-7} m^2. When a potential difference is applied to the ends of the wire, the current is 1.0 A.



If all the electrons within a cylinder of length L pass point X in 60 s, what is the value of L?

A 0.00015 m **B** 0.014 m **C** 0.025 m **D** 0.20 m

5. [JJC/H2/Prelims 2018/P1/18]

The current in a component is reduced uniformly from 100 mA to 20 mA over a period of 8.0 s.

What is the charge that flows during this time?

A 160 mC **B** 320 mC **C** 480 mC **D** 640 mC

6. [RVHS/H2/Prelims 2018/P1/18]

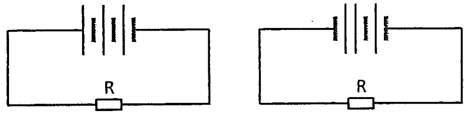
A uniform copper rod of cross-sectional area 8.0×10^{-6} m^2 has 8.5×10^{28} conduction electrons per cubic metre. Current flows through the rod when a potential difference is applied across it.

Given that the drift velocity of electrons in the rod is 2.3×10^{-5} m s^{-1}, what is the current in the rod?

A 0.25 A **B** 0.40 A **C** 2.5 A **D** 4.0 A

7. [VJC/H2/Prelims 2018/P1/20]

Three identical cells each having an e.m.f. of 1.5 V and a constant internal resistance of 2.0 Ω are connected in series with a 4.0 Ω resistor R, firstly as in circuit (i), and secondly as in circuit (ii).



circuit (i) **circuit (ii)**

What is the ratio $\frac{\text{power in R in circuit (i)}}{\text{power in R in circuit (ii)}}$?

A 9.0 **B** 7.2 **C** 5.4 **D** 3.0

8. [YJC/H2/Prelims 2018/P1/24]

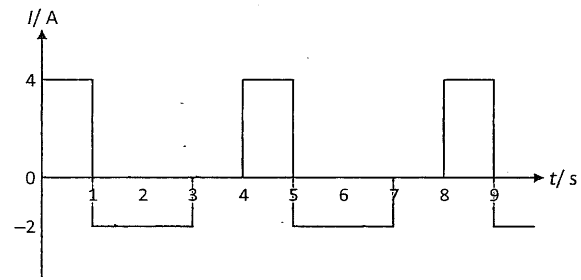
A wire has diameter 0.35 mm and the number density of charge carriers is 7.8×10^{28} m^{-3}. The current in the wire is 0.15 A.

What is the drift velocity of the charge carriers?

A 1.2×10^{-18} m s^{-1} **B** 3.1×10^{-11} m s^{-1} **C** 3.1×10^{-5} m s^{-1} **D** 1.2×10^{-4} m s^{-1}

9. **[ACJC/H2/Prelims 2018/P1/25]**

An alternating current with a rectangular waveform as shown below flows through an 11 Ω resistor.

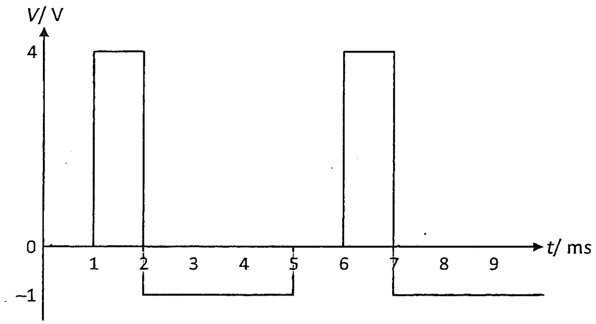


What is the average power dissipated by the resistor?

A 0 W **B** 44 W **C** 66 W **D** 88 W

10. **[AJC/H2/Prelims 2018/P1/24]**

The variation with time t of the voltage V of an alternating source applied across a 2.0 Ω resistor is shown below.

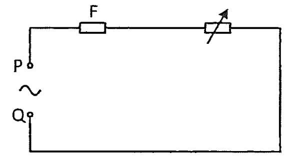


What is the power dissipated in the resistor?

A 0.98 W **B** 1.5 W **C** 1.9 W **D** 2.4 W

11. **[CJC/H2/Prelims 2018/P1/23]**

When an alternating power supply of 240 V r.m.s. is connected across PQ in the circuit shown below.



The fuse F breaks the circuit if the current in it just exceeds 13 A r.m.s.

When the alternating power supply is replaced with a 120 V d.c. source, an identical fuse breaks the circuit if the current just exceeds

A $\frac{13}{2}$ A **B** $\frac{13}{\sqrt{2}}$ A **C** 13 A **D** 26 A

12. **[CJC/H2/Prelims 2018/P1/24]**

An alternating power supply of root-mean-square voltage 4.0 V is connected across a resistive load such that the average power dissipated across it is P.

What is the d.c. voltage applied across the same load which will give rise to an average power dissipation of $3P$?

A 6.9 V **B** 8.5 V **C** 12 V **D** 17 V

13. **[DHS/H2/Prelims 2018/P1/26]**

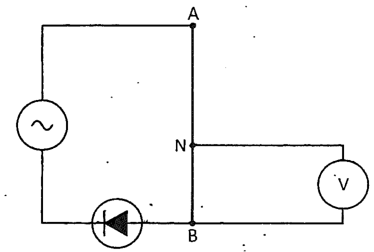
A sinusoidal alternating current of peak value I_0 passes through a resistor of resistance R. The mean power developed by the current in the resistor is P.

Another sinusoidal alternating current passes through a resistor of resistance $2R$. If the mean power developed by this current in it is $4P$, what is the root-mean-square value of this current?

A $0.7I_0$ **B** I_0 **C** $1.4I_0$ **D** $2.0I_0$

14. **[EJC/H2/Prelims 2018/P1/24 Modified]**

The uniform wire AB has length 1.0 m and resistance of 10 Ω. When NB is 40.0 cm, the a.c. voltmeter reads a steady r.m.s. voltage of 2.5 V.



What is the peak power provided by the supply?

A 0.63 W **B** 3.9 W **C** 7.8 W **D** 16 W

3. [YJC/H2/Prelims 2018/P3/7]

a. Domestic users in the United Kingdom are supplied with mains electricity at a root-mean- square voltage (r.m.s.) of 230 V.

i. State what is meant by root-mean-square voltage. [1]

ii. Calculate the peak power dissipated in a lamp connected to the mains supply when the r.m.s. current is 0.26 A. [2]

b. Another alternating voltage is shown in Fig. 7.1.

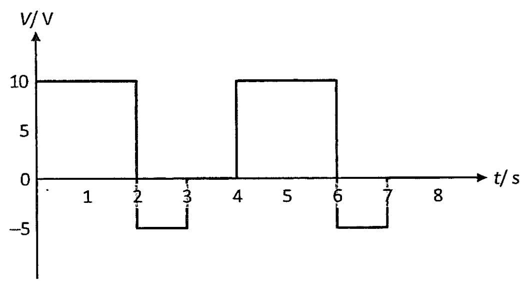


Fig. 7.1

Determine the root-mean-square voltage. [2]

[Total: 5]